

Orthogonal frequency division multiplexing phenomenology: recognition of canonical scatterers using flat spectra OFDM pulses

Gabriel Lellouch ✉, Amit Kumar Mishra, Michael Inggs

Department of Electrical Engineering, University of Cape Town, Cape Town, South Africa

✉ E-mail: gabriel.lellouch@gmail.com

ISSN 1751-8784

Received on 11th November 2014

Revised on 22nd June 2015

Accepted on 27th September 2015

doi: 10.1049/iet-rsn.2014.0469

www.ietdl.org

Abstract: In this study, the authors present a novel approach to characterise scattering centres from their radar return using specially designed waveforms. The widely accepted scattering model derived from the geometrical theory of diffraction is considered. They show how the use of flat spectrum orthogonal frequency division multiplexing pulses can help discriminating between the various scattering mechanisms. The technique, which assumes prior information about the target runs into two steps. The first step consists of designing pulses of appropriate length to mitigate intersymbol interference when several scatterers compose the target. In the second step, they calculate the spectrum of the received echo and estimate the slope of this spectrum, whose value is used to infer the scatterer type. Eventually, they assess the sensitivity of this approach with respect to additive white Gaussian noise in terms of the overall probability of correct classification and show that an integration technique can help improve the classification rate even in low signal-to-noise ratio conditions.

1 Introduction

A big part of the early developments in radar up to now has been to come up with favourable waveforms and signal processing techniques to detect the presence of a target, measure its range and radial velocity, localise its position [1, 2]. Ultimately, the goal of radar is to identify the target, especially in military applications. The generic procedure consists of collecting features of the target and later comparing them with a database to find the best match. In this, both high range resolution profiles [3] and synthetic aperture radar (SAR) images [4] have been used.

Such classic techniques suffer the same paradigm which was noted by Huynen in [5]: 'It may seem strange to say that the behaviour of a radar target as an object is treated with the utmost indifference in most radar systems applications; however, a search through the literature soon convinces us that this is indeed the case. The single object is usually considered as a 'point source' of scattering and an extended target as a distribution of point sources.' Indeed, techniques as recent as inverse synthetic aperture radar (ISAR) have built their theoretical framework upon the point target model. Although this approach reduces many difficulties related to the modelling of extended targets, its pertinence in the achievement of the final goal remains questionable. Hence, the realisation of this 'negligence' has urged Huynen to propose a 'phenomenological approach' instead. He exposed in [6] how the latter differs from the conventional approach which he calls the 'data processing oriented' approach. From a strictly data processing point of view, any data relating to the signal scattered off a target of interest is considered as potentially useful information regarding that target. He summarises the data processing people's philosophy as: 'the more data one can accumulate from a target, the more accurate the final identification is bound to be'. This simplistic approach engenders problems as the vast number of parameters that can characterise a set of targets soon overloads the data processing and procedures to discriminate between targets become cumbersome. Besides, the transient properties of the radar measurement phenomenon implies that for any new measurements involving new targets, new environments, new frequencies and so on, the whole procedure shall be repeated again. In contrast, the proposed approach was called phenomenological because it aims at keeping the primary phenomenon in continual focus of the observer's attention. When the data processor welcomes

any new piece of information to work towards the achievement of its objectives the phenomenologist tries to filter out only those data which relate to distinctive target behaviour. The benefice of this approach is twofold: more accurate discrimination capabilities obtained with less complex data processing.

In this work, we apply the phenomenological approach to orthogonal frequency division multiplexing (OFDM) radar. OFDM radar has become an area of active research because of the unique advantages the OFDM structure offers in the waveform design phase [7–10]. While Huynen inspected the target polarimetric properties, we focus on the frequency response. In [11], within the same scope of OFDM phenomenology we proposed a new framework to improve the recognition functionality in OFDM radar, when the targets radar cross section (RCS) are deterministic and known. In this work, we analyse the case of canonical scattering objects and show how we can benefit from the OFDM structure to identify the scatterer type. We develop a new framework that requires some prior information about the target. Therefore we recommend applying it in controlled environments, for example in civilian applications.

The remaining of the paper is organised as follows. In Section 2, we first introduce the scattering model used to describe the canonical scatterers. Then, we present the methodology of our synthetic experiment in Section 3. We state the problem, evoke the prior work and elaborate on our contribution. We also present our simulation setup and target model. Next, in Section 4, we review our technique to estimate the characteristics of the canonical scatterers. We motivate the use of flat spectrum OFDM pulses and stress the need to have short non-overlapping echoes in time. We wrap up this section with a statistical analysis and show that an integration technique can improve the probability of correct classification for the scatterer type especially as the level of noise increases. Section 5 concludes the paper.

2 Canonical scattering model

In recent years, several authors approached the problem of scattering centre characterisation for use in automatic target recognition, mostly in SAR applications [12, 13]. They grounded their research on the hypothesis that radar measurements of man-made targets are

dominated by the returns from isolated canonical scattering centres such as spheres, cylinders, corner reflectors and so on. Indeed, from the geometrical theory of diffraction (GTD) [14, 15], if the wavelength of the excitation is small relative to the object extent, then the backscattered electromagnetic field from an object consists of contributions from electrically isolated scattering centres and the resulting electric field is given by

$$E(f; \theta) = \sum_{m=1}^M E_m(f; \theta_m), \quad (1)$$

where $\theta^T = [\theta_1^T, \dots, \theta_M^T]$.

When the object lies in the far field, the GTD predicts that the scattering phenomenon follows an $(jf)^\alpha$ dependence, where α is a multiple of $1/2$, [16–18]. In our one-dimensional approach, the model is the same as used in [19], viz.

$$E_m(f) = A_m \left(j \frac{f}{f_c} \right)^{\alpha_m} \exp \left(\frac{-j4\pi f}{c} r_m \right). \quad (2)$$

The model parameter vector $\theta = \{A_m, r_m, \alpha_m\}_{m=1}^M$ characterises the M individual scattering centres. For each scatterer, A_m is a complex scalar that provides its magnitude and phase, r_m corresponds to its distance from the radar and α_m is a dimensionless coefficient that retains the geometry information. As we shall see, our focus is on a new method to retrieve α_m for each scatterer.

The scattering mechanisms are of two types: diffraction or reflection. The values for α in each of the five cases considered are summarised in Table 1.

This model generalises the conventional point scattering model obtained when $\alpha = 0$.

3 Methodology

Before discussing our contribution, we address the problem at stake and review the state-of-the-art. We then present our simulation setup and describe our target model.

Assuming that the target in the scene is composed of multiple scattering centres, our objective is to retrieve the characteristics of the scatterers. In other words, we aim to evaluate the vector of parameters θ .

So far, the authors have taken two different approaches to tackle the problem of scattering centre extraction. The first technique has been to compute the inverse discrete Fourier transform (IDFT) of the scattered signal to produce an impulse response of the target. The scatterers are then identified by localising the peaks of the estimated range profile [20]. The other technique has been to make use of a parametric model to describe the measured scattering behaviour [19]. The parameters describing the model are then estimated from the data. The second approach has a major advantage in that the range resolution is not bandwidth limited, rather, it is constrained by the parameter estimation error and the characteristics of the scatterers can be estimated. In [19], the authors present an approximate maximum-likelihood algorithm that aims at minimising the squared error $J(A, r, \alpha)$ defined by

$$J(A, r, \alpha) = \left| E_n - \sum_{m=1}^M A_m \left(j \frac{f}{f_c} \right)^{\alpha_m} \exp \left(\frac{-j4\pi f}{c} r_m \right) \right|, \quad (3)$$

Table 1 Parameter α and characteristic of the scattering centre

Scattering geometry	α
flat plate at normal incidence	1
singly curved surface (e.g. cylinder at broad side)	0.5
doubly curved surface (point scatterer)	0
edge	-0.5
corner	-1

where $E_n = E(f_n; \theta) + e_n$; f_n are the frequency samples within the bandwidth of interest and e_n are complex, zero mean, independent and identically distributed, Gaussian random variables with variance σ_n^2 .

The problem of parametric scattering centre information extraction is greatly simplified by our proposed work where we exploit OFDM signal used with a phenomenological intent. In essence, we apply conventional radar signal processing to estimate the range and the amplitude of the scatterers. Our contribution lies in the combination of flat spectrum OFDM pulses on transmit with an algorithm that estimates the slope of the received echo spectrum. This slope relates directly to the amplitude variation caused by the α_m coefficient in (2). Hence, our statistical analysis, presented in Section 4.4 only concerns the type of the scatterer, viz. α_m . With this study, our intention is to propose a new framework to this problem of recognising the scatterer type. The study of better algorithms forms the part of the future work. Note that a thorough analysis concerning the Cramer-Rao lower bound for the variance achievable by any unbiased estimator for either three parameters has been performed in [19]. It is not the aim here to repeat it.

3.1 Simulation setup

In our approach, the successive steps undergone by the radar signal are summarised as follows. The transmitted signal $s(t)$ propagates away from the radar antenna and interacts with the scatterer whose deterministic reflectivity function is given by $h_m(t)$. The scattered signal $z_m(t)$ is then received at the radar antenna. Additive white Gaussian noise (AWGN) adds onto the signal and results in the compound signal $y_m(t)$. If the conventional matched-filtering process is applied, we eventually obtain the compressed signal $x_m(t)$. In this analysis, we assume that the transmitted signal is analytical. The spectrum of the received baseband signal is given as in [21] by

$$Z_{bm}(f) = H_m(f + f_c) S_b(f), \quad (4)$$

where $S_b(f)$ is the spectrum of the baseband pulse. If $y_{bm}(t)$ and $s_b(t)$ are, respectively, the baseband compound signal and the baseband transmitted pulse, the output of the matched-filter $x_m(t)$ is given by

$$x_m(t) = y_{bm}(t) * s_b^*(t). \quad (5)$$

We use a simple energy detector to detect the scatterers. An estimate of the range $\hat{r}_m$ is given by the peak position on the range axis and an estimate of the scattering coefficient is given by the value $x(\hat{r}_m)$, where $\hat{r}_m = 2\hat{r}_m/c$.

3.2 Target modelling

In light of (2), we simulate the target in the frequency domain. The target impulse response is then obtained after applying an inverse Fourier transform (IFT) as in (6) and the Dirac-delta at the scatterers' positions are generated. Note that the reflectivity spectrum H can be used interchangeably with the backscattered electromagnetic field E as defined in (1). When M scatterers compose the target, $H = \sum_{m=1}^M H_m$.

$$h(t) \exp(-j2\pi f_c t) = \text{IFT}(H(f + f_c)). \quad (6)$$

In the discrete-time formulation, the reflectivity spectrum is sampled at a rate such that the scatterers appear at their true position in the target impulse response, so as to avoid aliasing. The interval of frequencies between two samples is referred to as Δf , not to be confused with Δf which we introduce later to characterise the OFDM subcarrier spacing. In vector format, the reflectivity spectrum is $\mathbf{h}_f$ and the target impulse response resulting from applying an IDFT on $\mathbf{h}_f$ is $\mathbf{h}_t$. Since we work on

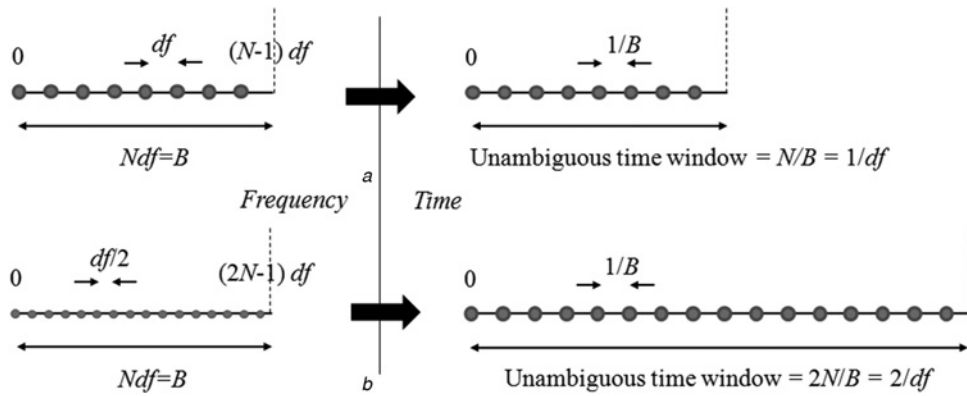


Fig. 1 Effect of oversampling in the frequency domain. In (b) we use twice more samples in the frequency domain than in (a) while maintaining the same bandwidth. Consequence in the time domain is a vector twice as long. Sampling period is not modified

the reflectivity spectrum at radio frequency (RF), $\mathbf{h}_t$ absorbs the complex exponential from (6). Note that in principle the impulse response can be infinite due to the effects of multiple reflections of the radar energy from different segments of the extended target, however, such effects of multiple reflections become vanishingly small after some time duration on the order of the waveform transit time across the target, hence an approximation of $\mathbf{h}_t$ based upon a finite size vector is sufficient for numerical simulations.

In this analysis, we consider two cases: a pointed and an extended target. The minimum distance between any two consecutive scatterers in the first case is smaller than in the second case. Consequences on the pulse design are given in Section 4.1. Both targets consist of five scatterers. In the small target case, the scatterers are positioned at the ranges: 4997, 5000, 5006, 5010 and 5021 m, hence the minimum distance between two consecutive scatterers is $\delta R_{\min} = 3$ m and the target extent is $\Delta R = 24$ m. In the extended target case, the scatterers are positioned at the ranges: 5000, 5030, 5052, 5090 and 5120 m, hence the minimum distance between two consecutive scatterers is 22 m and the target extent is 120 m. In both cases, the first and the closest scatterer have the type $\alpha = 1$, the second scatterer has the type $\alpha = 0.5$, the third scatterer has the type $\alpha = 0$, the fourth scatterer has the type $\alpha = -0.5$ and the fifth and the furthest scatterer have the type $\alpha = -1$.

The relationship between time and frequency imposed by the Fourier transform implies an unambiguous range $R_{ua} = c/2df$. Thus, df must be chosen to comply with the target extent. Fig. 1 illustrates the principle whereby oversampling in the frequency domain increases the length of the unambiguous time window. In our simulations, we choose for the small target case, $df = 5.56$ MHz which results in $R_{ua} = 27$ m while in the extended target case, $df = 1.11$ MHz and $R_{ua} = 135$ m. These values are summarised in Table 2. We observe that the unambiguous window characterised by its boundaries $R_{win \text{ start}}$ and $R_{win \text{ stop}}$ includes all the scatterers.

For reasons that will become clear in Section 4.1, we considered two values for the bandwidth as shown in Fig. 2. In the extended target case, we used $B = 300$ MHz and in the small target case we used $B = 1500$ MHz.

4 Estimation technique

In this section, we review our proposed framework for the recognition of canonical scatterers. In particular, we motivate the use of flat spectrum OFDM pulses to estimate the α_m coefficients. We first explain our methodology to design the pulses, then address the generation of our synthetic data and present our algorithm for the parameter estimation problem. We wrap up with a statistical analysis to assess the behaviour of our technique under AWGN conditions.

4.1 Pulse design

Our approach assumes to have some prior information on the relative distance between the scatterers. The pulse is thus designed to be no longer than the minimum distance between two consecutive scatterers. This consideration reduces the risk of intersymbol interference (ISI) when we apply our algorithm.

4.1.1 Subcarrier spacing: Figs. 3a and d show two different designs intended for two different target configurations.

In the first case, the pulse length shall be no longer than 3 m, equivalently $0.02 \mu\text{s}$ in time, while in the second case it shall be no longer than 22 m, equivalently $0.15 \mu\text{s}$. The orthogonality property of the OFDM structure implies that the subcarrier spacing shall be at least 50 and 6.8 MHz, respectively. To keep things simple, we use $\Delta f = 10$ MHz in the second case and we consider the same number of subcarriers $N = 30$ for both, so that we use the same number of samples to estimate the slope of the spectrum in Section 4.3. As a result, the pulse bandwidth in the small target case will have to be five times larger than in the extended target case. These choices are summarised in Table 3. The pulse extent in range is found to be $c/2\Delta f$. Only one symbol composes the OFDM pulse and in both cases, the same set of random phase codes and the same set of weights are used to generate the OFDM pulse, hence the pulse in Fig. 3d is a stretched version of the pulse in Fig. 3a. Note that other arrangements of the phase codes could be used if less variation of the time domain envelope is needed [10].

4.1.2 Weights: Importantly, the subcarriers composing the pulse have the same weight. Hence, the spectrum of the transmitted pulse $S_b(f)$ in (4) is flat. As a matter of fact, the reflectivity spectrum of the scatterer $H_m(f)$ at RF reflects directly onto the spectrum of the received echo $Z_{bm}(f)$. It is this property that we use to retrieve the type of scattering in Section 4.3.

As pointed out before, this technique requires prior information regarding the position of the scatterers on the target and their relative separation in terms of range. Hence, we reckon it shall be used in controlled environments (e.g. in civilian applications). Importantly, high energy OFDM pulses shall be used in order to infer suitable signal-to-noise ratio (SNR) characteristics as we shall see in Section 4.4.

Table 2 Parameters involved in the design of the experiment

	Small target	Extended target
$\delta R_{\min}$	3 m	22 m
ΔR	24 m	120 m
df	5.56 MHz	1.11 MHz
R_{ua}	27 m	135 m
$R_{win \text{ start}}$	4995 m	4995 m
$R_{win \text{ stop}}$	5022 m	5130 m

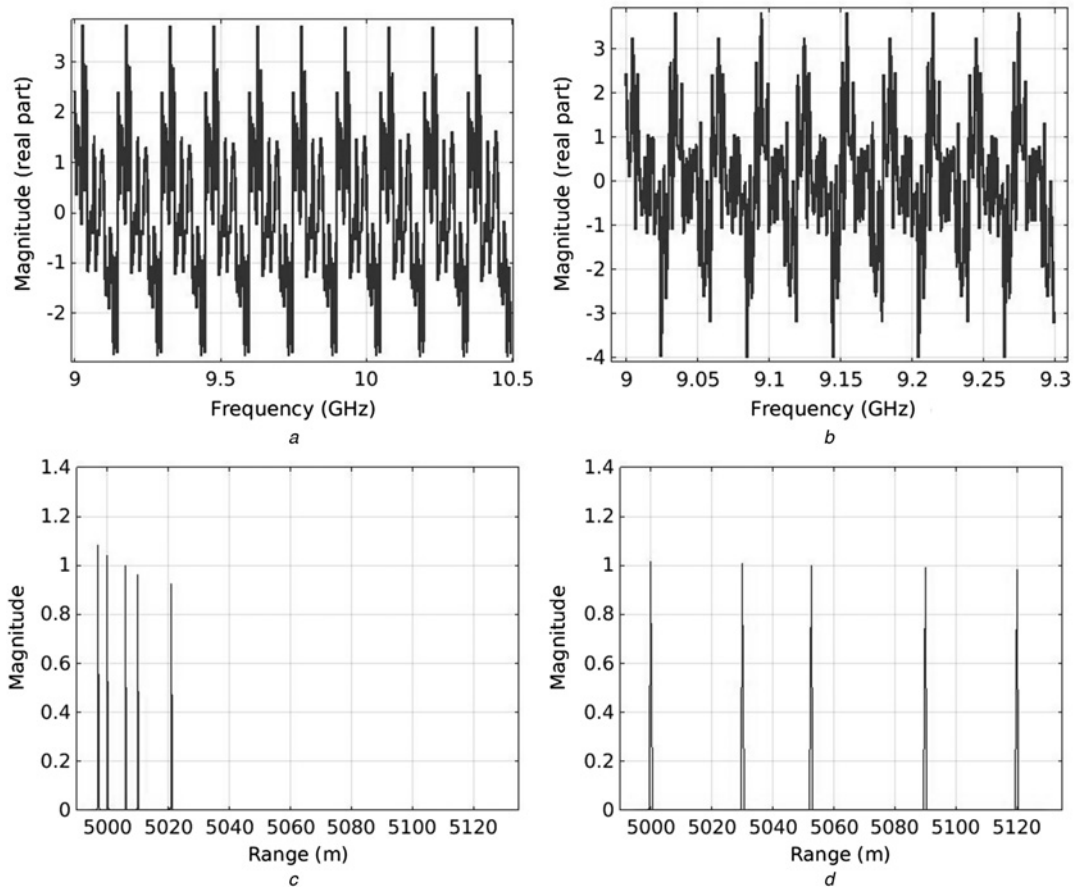


Fig. 2 Target modelling. In (a) and (b) we display the reflectivity spectra of the small and long targets, over 1500 and 300 MHz, respectively. In (c) and (d) we show the corresponding impulse responses

- a Reflectivity spectrum, small target
- b Reflectivity spectrum, long target
- c Impulse response, small target
- d Impulse response, long target

4.2 Synthetic data generation

To generate our set of synthetic data we make use of the convolution theorem and (4). As a result, the received echo z_b can be expressed as

$$z_b = h_t * s_b. \quad (7)$$

In Figs. 3b and e, the absolute value of the received signal at baseband is presented for the small and extended target cases. As expected from the prior design, pulses in the small target case are five times shorter than in the extended target case. In either case, the impulse response vector h_t and the pulse vector s_b have the same bandwidth B , hence, the sampling rate of the resulting vector z_b is also B .

4.3 Algorithm

To retrieve the type of scattering, we first evaluate the spectra of the received echoes z_{bm} using a discrete Fourier transform (DFT) and then assess the behaviour of each spectrum with respect to frequency. Lastly, we refer to the cases presented in Table 1 to link the parameter α_m and the scatterer type. We summarise our algorithm in Fig. 4.

Importantly, the magnitude of the spectrum shall be expressed in dB, as well as the frequency axis to reveal a linear expression of the form $|Z_{bm}|_{dB} \propto \alpha_m(f/f_c)_{dB}$. The curve fitting is thus done with a first order polynomial. To mitigate the magnitude glitch we restrict the fitting to the inner samples and excludes both two samples on the edges of the spectra as shown in Figs. 5a and b. For the last

step, we suggest the following basic decision rule. We suppose five ± 0.24 windows centred around the five values of α , respectively. Having assessed the slope of the spectrum, which in turn provides $\hat{\alpha}_m$, we simply attribute to our scatterer the value of α corresponding to the window that contains $\hat{\alpha}_m$.

4.4 Statistical analysis

To assess the performance of our technique we employ a statistical approach. The idea is to characterise the slope estimate when AWGN adds up onto our received echoes. We define our signal model as follows: first, we assess the energy of the individual echoes $\varepsilon = z_{bm}^H z_{bm}$, which we assume to be equivalent to the energy of the pulse [To calculate the energy of the pulse we consider the critical sampling rate without oversampling, hence the sampling frequency is $f_s = B$.] s_b since the impulse response of each scatterer is approximately a discrete Dirac with amplitude 1 as shown in Figs. 2c and d. Hence we work with energy-normalised pulses, $\varepsilon = 1$; $\cdot^H$ denotes the Hermitian transposition and we use the convention of column vectors. Then, we calculate the energy that shall have the noise vector of equal length, for a specific SNR, viz. $n_p = \varepsilon/\text{SNR}$ where SNR is in linear scale and we generate the vector of complex AWGN as in the following equation

$$n = \sqrt{\frac{n_p}{2N}}(n_1 + jn_Q), \quad y_b = z_b + n. \quad (8)$$

n_1 and n_Q are vectors of equal length as z_b whose elements are drawn

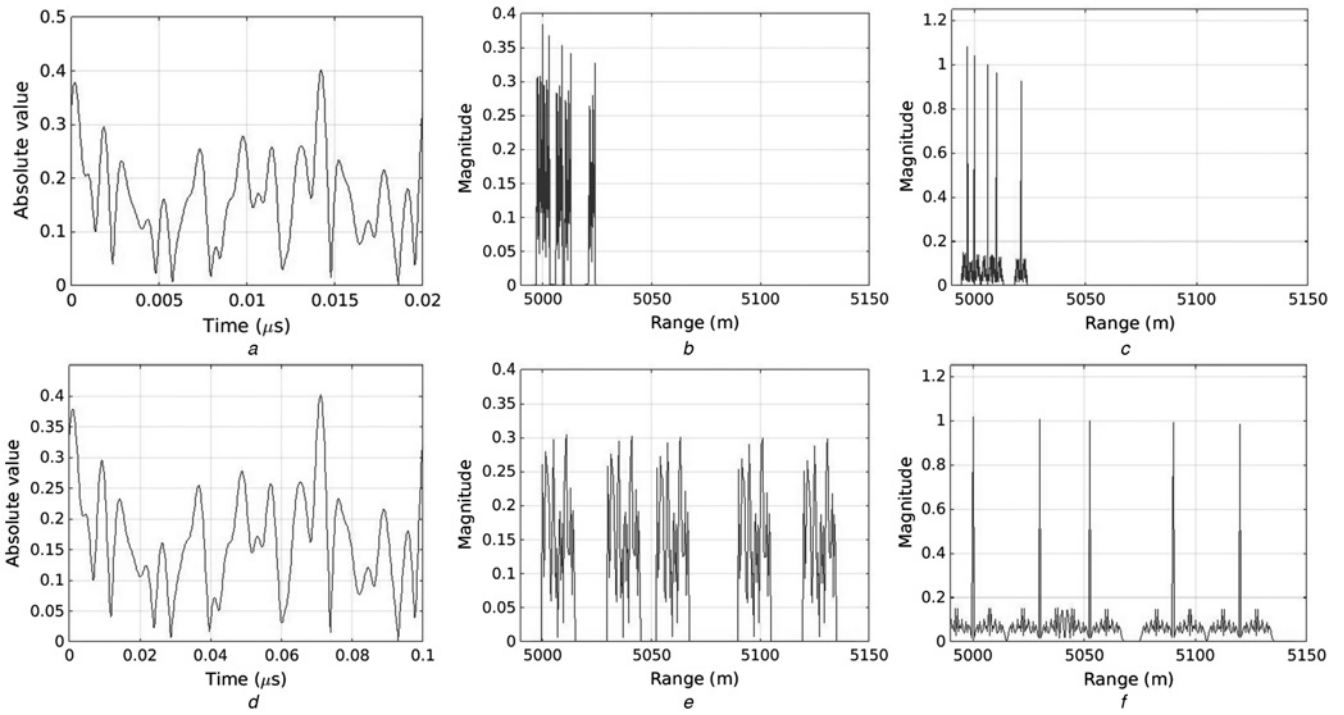


Fig. 3 Radar pulse, received echoes and compressed echoes for (a)–(c) small and (e)–(g) extended targets. In (a) $t_p = 0.02 \mu s$ and in (d) $t_p = 0.1 \mu s$. In both cases, the OFDM pulse is generated with the same set of phase codes as well as the same set of weights and the pulse length is no longer than the minimum distance between two consecutive scatterers. Sampling rate of the received echo z_b in range is $c/2B$, that is in (b) $0.1 m$ and in (e) $0.5 m$. Both radar pulses have been plotted with an oversampling factor of 20

a Small target pulse
b Small target echoes
c Small target compressed echoes
d Extended target pulse
e Extended target echoes
f Extended target compressed echoes

Table 3 Parameters involved in the design of the OFDM pulse

	Small target	Extended target
N	30	30
B	1500 MHz	300 MHz
Δf	50 MHz	10 MHz
$c/2\Delta f$	3 m	15 m

from a normal distribution with mean 0 and variance 1. The compound signal y_b corresponds to the vector of noisy echoes and y_{bm} is the noisy echo associated to scatterer m subject to the specified SNR; $j = \sqrt{-1}$.

Algorithm 1

- 1: Apply a matched filter on the vector of received echoes z_b
- 2: Find the peaks / detections
- 3: Verify that the spacing between two consecutive detections is more than the pulse length
- 4: **for** Each detection m **do**
- 5: Read the position of the peak and its value to estimate the scatterer range $\hat{r}_m$ and amplitude $|\hat{A}_m|$
- 6: Extract the received echo vector z_{bm}
- 7: Apply a DFT on z_{bm}
- 8: Convert the spectrum from linear to dB, in the form $10 \times \log(\text{Magnitude})$ versus $10 \times \log(f/f_c)$
- 9: Run a curve fitting of order one on this dB scale spectrum and evaluate the slope of the line to find $\hat{\alpha}_m$.
- 10: Deduce the type of scattering with the help of table I
- 11: **end for**

Fig. 4 Retrieving the scatterer type

We now present our results when y_b replaces z_b in the algorithm (see Fig. 4). The output of the matched filter process, presented in Figs. 3c–f for both the small and extended target cases is used to retrieve the number of scatterers M and estimate both $|\hat{A}_m|$ and $\hat{r}_m$. As mentioned earlier our emphasis is on the classification of the type of scatterer. Hence, our subsequent analyses focus on the slope estimation and scatterer type retrieval. In particular, we run our algorithm from steps 7 to 10 on y_{bm} and see how the estimate $\hat{\alpha}_m$ behaves.

4.4.1 Overall probability of correct classification: For every SNR level considered, we generate $M_r = 5000$ realisations of the noise. Hence, as we iterate the previous algorithm, we retrieve 5000 slope values for each scatterer. Our intention is to inspect the degradation of the canonical scatterer classification, which we formulate in terms of the overall probability of correct classification P_{cc} , as the SNR decreases. In particular, for every realisation and for a given scatterer we declare a correct classification when the estimated slope falls within the appropriate interval, as described earlier. For the overall probability of correct classification we use the same notation as in [22] and express it in terms of the probability of correct classification for a given scatterer type i as

$$P_{cc} = \frac{1}{C} \sum_{i=1}^C P_{cc,i}, \quad (9)$$

where C is the number of classes considered for the canonical scatterer. The index i links to the possible discrete values for α , $\{-1, -0.5, 0, 0.5, 1\}$. For low SNR values, we will see that the integration of many noisy spectra from the same scatterer can enhance drastically the classification.

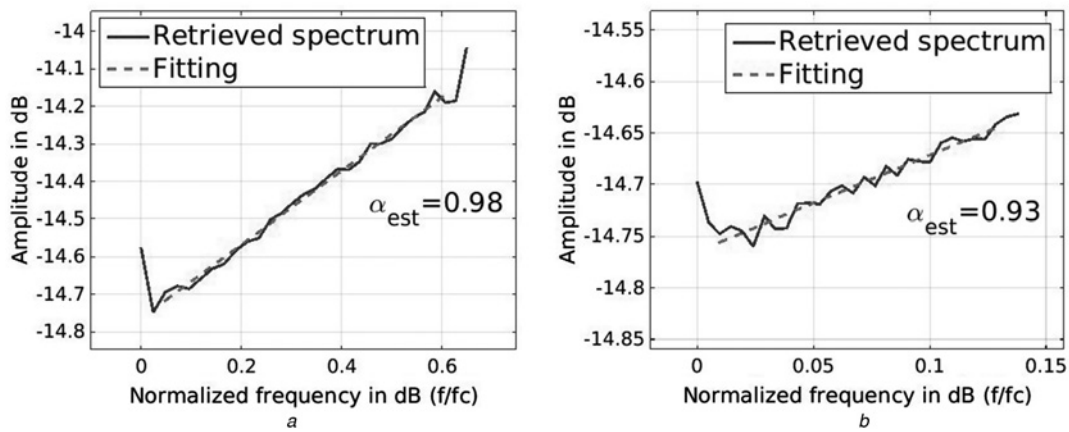


Fig. 5 Spectra of the received echoes when the bandwidth of the transmitted pulse was $B = 1500$ and 300 MHz. (a) Small target case and (b) extended target case

a Scatterer 1, $\alpha_1 = 1$, $B = 1500$ MHz
b Scatterer 1, $\alpha_1 = 1$, $B = 300$ MHz

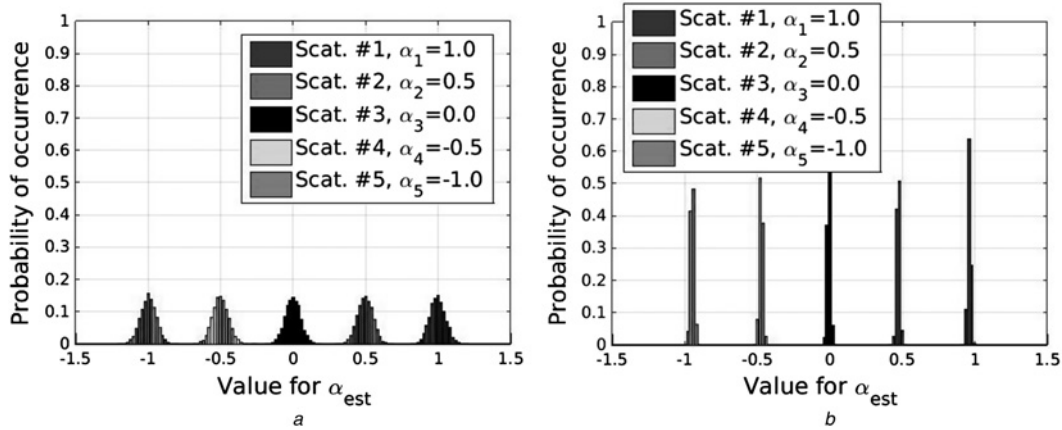


Fig. 6 Histograms of the slope estimates for both cases, (a) $B = 300$ MHz and (b) $B = 1500$ MHz when the SNR is 50 dB. Values in the individual histograms have been normalised so that the total sum adds up to 1

a $B = 300$ MHz
b $B = 1500$ MHz

4.4.2 Low noise level, $SNR = 50$ dB: For high SNR values, the probability density function (PDF) of the slope for each scatterer is well represented by a Gaussian distribution centred around the true value, as seen in Fig. 6a, for the case $B = 300$ MHz. In the other case, $B = 1500$ MHz, the peak happens to be slightly shifted for $\alpha = -1, -0.5, 0.5$ and 1 as observed in Fig. 6b. The mean and standard deviation values for all distributions are summarised in Table 4. Comparing the results for both bandwidths, we foresee that the standard deviation is inversely proportional to the relative bandwidth. Indeed, we can find that the standard deviation in the case $B = 1500$ MHz is about five times smaller than in the case $B = 300$ MHz.

Since we are interested in retrieving correctly the parameter α , rather than finding the exact slope value, the small bias (0.05 for $\alpha = -1$ represents a shift of 10%) is not detrimental with the choice of interval considered ± 0.24 . Since all slope estimates fall within

the appropriate window, we see that in either cases $P_{cc_i} = 100\%$ for all scatterer types. With our prior definition of the overall probability of correct classification, we find that $P_{cc} = 100\%$.

4.4.3 Overall probability of correct classification versus SNR: As the SNR decreases, the PDFs from Fig. 6 flatten and widen. Consequently, the number of slope estimates falling within the appropriate window decreases, so do the probabilities of correct classification P_{cc_i} . Hence, the overall probability of classification drops accordingly. In Fig. 7, we plotted P_{cc} as a function of SNR. Our findings seem to agree with the results presented in [19]. As we consider smaller relative bandwidth, the curve shifts to the right.

4.4.4 Integration of several pulses to improve the classification: One technique commonly used in radar when the probability of detection is poor with a single pulse is to integrate

Table 4 Mean value and standard deviation of the slope distribution for all five scatterer types in both cases, $B = 300$ MHz and $B = 1500$ MHz

B		Scatterer 1	Scatterer 2	Scatterer 3	Scatterer 4	Scatterer 5
300	mean	0.996	0.497	-0.002	-0.501	-0.999
	std	0.054	0.053	0.053	0.054	0.054
1500	mean	0.962	0.471	-0.007	-0.473	-0.949
	std	0.011	0.011	0.011	0.012	0.012

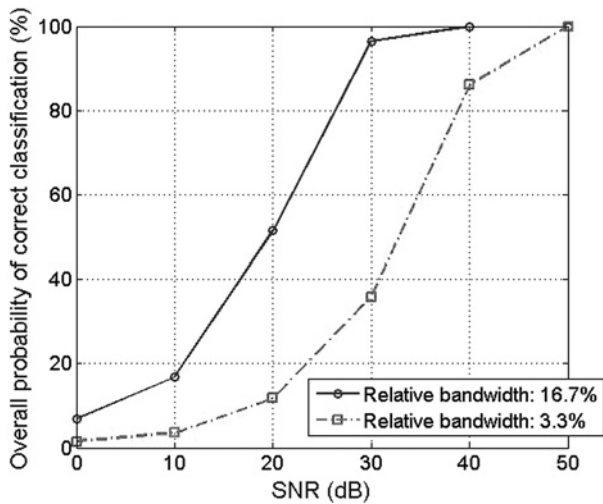


Fig. 7 Overall probability of correct classification versus SNR using $f_c = 9$ GHz, $N = 30$ subcarriers and two cases for the bandwidth, $B = 300$ and 1500 MHz. Window extent considered to compute the probability of correct classification for a given scatterer type i is $\alpha \pm 0.24$ where $\alpha = \{-1, -0.5, 0, 0.5, 1\}$

over several pulses. The randomness of the noise averages out, hence the higher the number of pulses used in the integration, the more significant the contrast between the target and the noise floor and the detection is improved. In the present case, we suggest a similar technique to improve the individual and then the overall probability of correct classification. Instead of summing up the individual echoes, we propose to integrate coherently their noisy spectra. Naturally, we use echoes that correspond to the same scatterer in the integration. We assess the behaviour of that approach with Monte Carlo simulations where we employ the same data set as used in Section 4.4.3 to generate Fig. 7. We define the number of pulses for integration as N_p and for several values of N_p , we select at random N_p noisy echoes in the pool of 5000. Thus, if $\mathbf{d}_m(i_k) = \text{DFT}(\mathbf{y}_{bm}(i_k))$ is the vector of complex

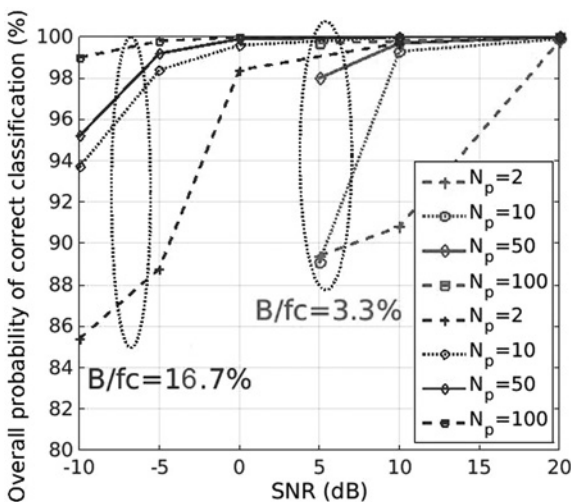


Fig. 8 Overall probability of correct classification versus SNR using $f_c = 9$ GHz, $N = 30$ subcarriers and two cases for the bandwidth, $B = 300$ MHz (relative bandwidth 3.3%) and $B = 1500$ MHz (relative bandwidth 16.7%) resulting from the integration of several spectra. Window extent considered to compute the probability of correct classification for a given scatterer type i is $\alpha \pm 0.24$ where $\alpha = \{-1, -0.5, 0, 0.5, 1\}$. For both cases, we report only those relevant results. In the low relative bandwidth case, performance improvement is happening from SNR = 5 dB while it happens from SNR = -10 dB in the high relative bandwidth case

spectrum corresponding to scatterer m for one realisation i_k , the vector of cumulated spectra is given as in the following equation

$$\mathbf{d}_m = \sum_{k=1}^{N_p} \mathbf{d}_m(i_k). \quad (10)$$

We then apply steps 8–10 of our algorithm on $\mathbf{d}_m$. We repeat this procedure $M = 5000$ times (this results in equally as many vectors of indices $[i_1, \dots, i_{N_p}]$) and visualise the results in histograms as shown in Fig. 6. As opposed to the single pulse case, the histograms do not produce clear Gaussian-like distributions especially for low SNR values. In Fig. 8, we display only those relevant results to attest the merit of our technique and the improvement in comparison to the single pulse case from Fig. 7. We observe that satisfactory values of P_{cc} are achieved with reasonable values of N_p , for instance $P_{cc} = 98\%$ for $N_p = 50$ with SNR = 5 dB for the low relative bandwidth case. For the high relative bandwidth case, satisfactory results are obtained from SNR = -10 dB onward: $P_{cc} = 99\%$ for $N_p = 100$ with SNR = -10 dB, for example. Hence, in both cases, a gain of about 40 dB is obtained in comparison to the single pulse case for the same performance.

5 Conclusion

We have presented a new framework for scattering centres extraction using flat spectrum OFDM pulses. Our emphasis was put on retrieving the scatterer type. The underlying principle of our method relies on estimating the slope of the OFDM echo spectra. We stressed the need to have separable echoes in order to prevent from any ISI to occur. We showed how prior information over the scatterers' relative distance enabled the design of OFDM pulses to cope with that requirement. Next, we performed a statistical analysis to characterise the slope estimates when the echoes are subject to different SNR levels for two different bandwidth cases. We demonstrated the merit of integrating over several echo spectra to improve the slope estimate in low SNR conditions which in turn improves the scatterer type characterisation. We observed that for a given number of subcarriers, the larger the relative bandwidth of the OFDM pulse, the better the performance of the classification with low SNR levels. Lastly, directions for future work include computational analysis and sensitive analysis of the parameters as well as the study of better algorithms.

6 References

- Levanon, N.: 'Stepped-frequency pulse-train radar signal', *IEE Proc., Radar Sonar Navig.*, 2002, **149**, (6), pp. 297–309
- Berizzi, F., Martorella, M., Bernabo, M.: 'A range profiling technique for synthetic wideband radar', *IET Radar Sonar Navig.*, 2008, **2**, (5), pp. 334–350
- Boulay, T., Lagoutte, J., Mohammad-Djafari, A., et al.: 'A Fuzzy-Logic based non cooperative target recognition'. Eighth Int. Conf. on Signal Image Technology and Internet Based Systems (SITIS), 2012, **2012**, pp. 410–415
- Martorella, M., Giusti, E., Demi, L., et al.: 'Target recognition by means of polarimetric ISAR images', *IEEE Trans. Aerosp. Electron. Syst.*, 2011, **47**, (1), pp. 225–239
- Huynen, J.R.: 'Phenomenological theory of radar targets'. Ph.D. dissertation, Drukkerij Bronder-Offset NV, 1970
- Huynen, J.R.: 'A revisit of the phenomenological approach with applications to radar target decomposition'. Tech. Rep., DTIC Document, 1982
- Lellouch, G., Mishra, A.K.: 'Multi-carrier based radar signal optimization using genetic algorithm'. Proc. Third Int. Conf. on Soft Computing for Problem Solving, 2014, pp. 525–534
- Lellouch, G., Mishra, A., Inggs, M.: 'Impact of the Doppler modulation on the range and Doppler processing in OFDM radar'. Radar Conf., 2014. RADAR '14, 2014, pp. 803–808
- Lellouch, G., Mishra, A., Inggs, M.: 'Processing alternatives in OFDM radar'. Int. Radar Conf. (Radar), 2014, 2014, pp. 1–6
- Lellouch, G., Mishra, A., Inggs, M.: 'Design of OFDM radar pulses using genetic algorithm based techniques', *IEEE Trans. Aerosp. Electron. Syst.* (in press)
- Lellouch, G., Mishra, A., Inggs, M.: 'OFDM phenomenology: radar technique combining genetic algorithm based pulse design and energy detector for target recognition', *IET Trans. Radar Sonar Navig.* 2015, DOI: 10.1049/iet-rsn.2014.0470
- Potter, L.C., Moses, R.L.: 'Attributed scattering centers for SAR ATR', *IEEE Trans. Image Process.*, 1997, **6**, (1), pp. 79–91

- 13 Akyildiz, Y., Moses, R.L.: 'Scattering center model for SAR imagery'. Int. Society for Optics and Photonics Remote Sensing, 1999, pp. 76–85
- 14 Keller, J.B.: 'Geometrical theory of diffraction', *JOSA*, 1962, **52**, (2), pp. 116–130
- 15 Kouyoumjian, R.G., Pathak, P.H.: 'A uniform geometrical theory of diffraction for an edge in a perfectly conducting surface', *Proc. IEEE*, 1974, **62**, (11), pp. 1448–1461
- 16 Plonus, M., Williams, R., Wang, S.: 'Radar cross section of curved plates using geometrical and physical diffraction techniques', *IEEE Trans. Antennas Propag.*, 1978, **26**, pp. 488–493
- 17 Pathak, P.H.: 'Techniques for high-frequency problems', in (Lo, Y.T., Lee, S.W., Eds.): 'Antenna handbook' (Springer, 1988), pp. 195–311
- 18 Balanis, C.A.: 'Advanced engineering electromagnetics' (Wiley New York, 1989), vol. 20
- 19 Potter, L.C., Chiang, D.-M., Carriere, R., *et al.*: 'A GTD-based parametric model for radar scattering', *IEEE Trans. Antennas Propag.*, 1995, **43**, (10), pp. 1058–1067
- 20 Kennaugh, E., Moffatt, D.: 'Transient and impulse response approximations', *Proc. IEEE*, 1965, **53**, (8), pp. 893–901
- 21 Wilkinson, A., Lord, R., Inggs, M.: 'Stepped-frequency processing by reconstruction of target reflectivity spectrum'. Proc. of the 1998 South African Symp. on Communications and Signal Processing, 1998. COMSIG '98, September 1998, pp. 101–104
- 22 Mishra, A.K., Mulgrew, B.: 'Automatic target recognition', *Encycl. Aerosp. Eng.*, 2010, DOI: 10.1002/9780470686652.eae281